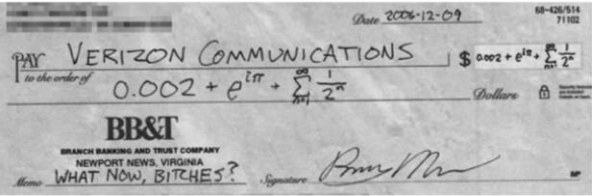
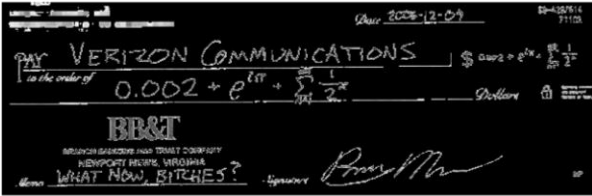


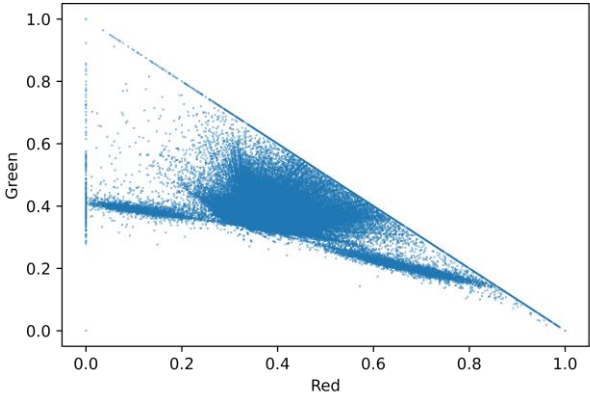
Original Grayscale



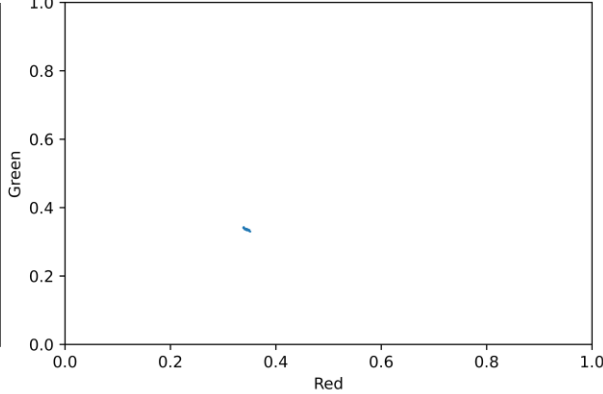
Threshold at 125



Normalized Chromaticity Coordinates of RGB Image



Normalized Chromaticity Coordinates of ROI



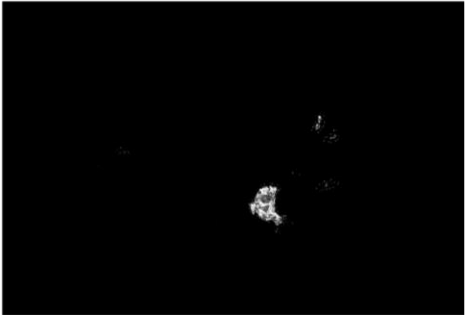
Original RGB Image



Parametric



Non-Parametric



Feature Extraction: Image Segmentation (Pre-Machine Learning)

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Objectives

- To isolate details of a grayscale image using segmentation by thresholding
- To isolate a region of interest in an RGB image using both parametric and non-parametric probability distribution estimation

Segmentation by Thresholding

Suppose we have an image in grayscale such as shown in Fig 1. We can represent it as a 2D matrix which we can manipulate easily. Suppose further that we want to isolate details of this grayscale object that has pixel values that are distinct. In this case, we want to isolate the handwriting in the figure which has darker pixel values than the background. How do we isolate the handwriting in this grayscale image of a check?

The answer is we introduce a threshold!

When a pixel has a value greater than the threshold (i.e. if it is bright) we set its pixel value to a flat 0 as to distinguish it as unimportant. When a pixel has a value lower than the threshold, we invert its value (i.e. if its value is 0/1 we convert it to 255/244) in order to see the handwriting clearly.

We can set the threshold to other values, but in this particular image the pixel value 125 seems to yield the best results. Note that this technique works if the Region of Interest (ROI) is distinct from the background when in grayscale.

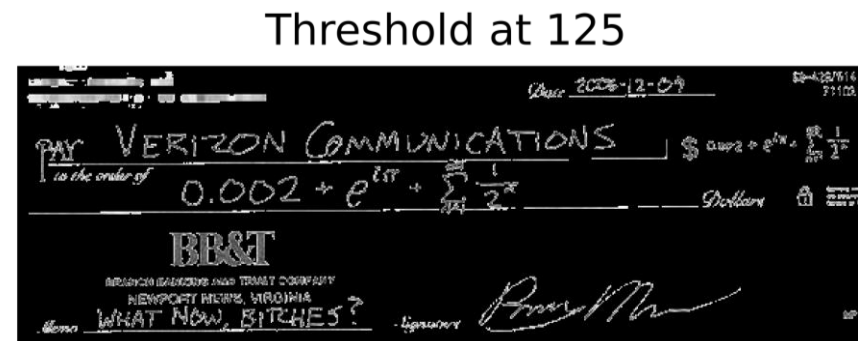
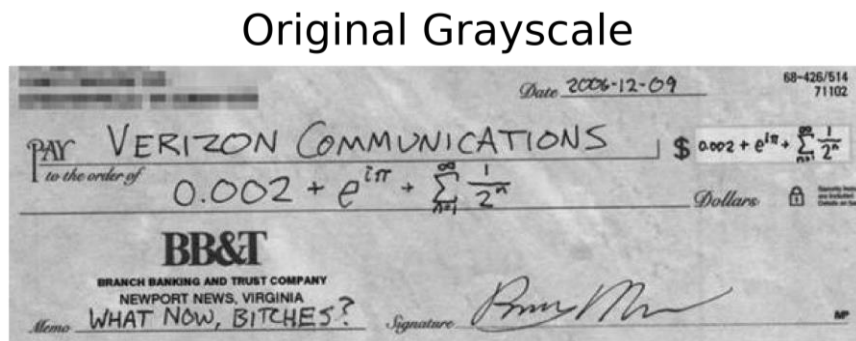


Fig 1. Grayscale image segmented by setting all pixels with value greater than 125 to 0

RGB and NCC Space

The previous method however, won't work for most RGB images since the details won't be distinct in grayscale. Objects captured by images will have different shading variations. A red pixel could appear not just as "redder", but also brighter or darker. There can also be reflections from light sources that will introduce a region of white light.

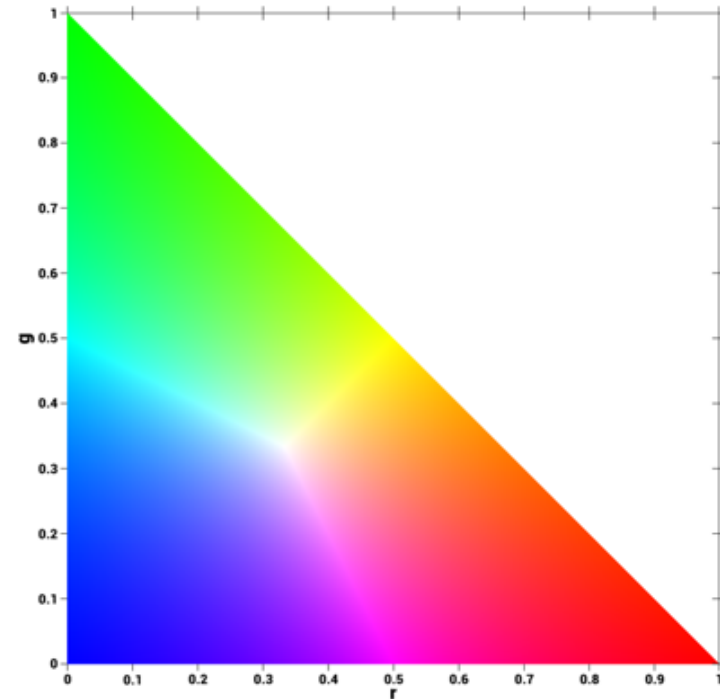


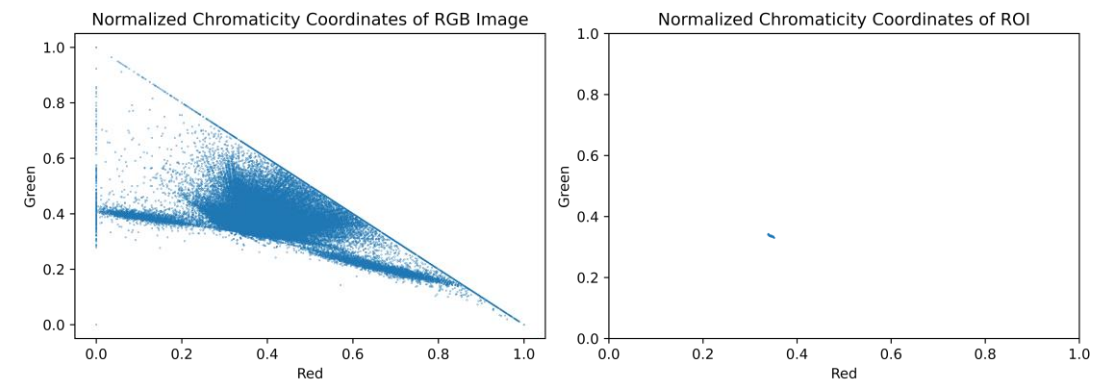
Fig 2. Normalized Chromaticity Coordinates [2]

For this reason, it is better to represent images not in RGB space but by something that can separate pure color (chromaticity) and brightness information. One such color space commonly used in segmentation is the **NCC (Normalized Chromaticity Coordinates)**.



Fig 3. RGB image (right) and ROI (left)

Here we have an image of Cuphead (right) running away from the devil with his brother Mugman (left) after losing a game of craps (don't bet away your soul kids). We select the ROI to be a cropped image of Cuphead's shorts! Let's plot them in NCC space!



We can see that the ROI is only centered at red values in NCC space, as it should be! Moreover, blue pixels are scarce – just as what is seen in the image. Now we have a histogram!

Parametric Segmentation

Colored images can be segmented by determining the probability that a pixel belong to the color of interest. We assume a Gaussian distribution along red and green in the NCC space independently. The probability that a pixel with chromaticity r belongs to the ROI is then

$$\rho(r) = \frac{1}{\sigma_r \sqrt{2\pi}} \exp \left(-\frac{(r - \mu_r)^2}{2\sigma_r^2} \right)$$

Where the mean and standard deviation of the RGB ROI image is taken. A similar equation is taken for g , and the joint probability is a product of both probabilities. What we will get is a 2D array of probabilities of each pixel's similarity to the ROI as shown in Fig 3.



Fig 4. Parametric segmentation of ROI assuming a Gaussian

There are artifacts of red in Mugman's blue pants because the game is entirely water colored. The redness of the canvas seeps through and the algorithm detects it.

This method is highly dependent on the fit of the PDF function assumed. The Gaussian is good but it can include unwanted colors with chromaticity values near the ROI depending on the image (like in mine where some canvas information was included!). Nevertheless, it's still good. We can teach another algorithm to choose only maximum intensities to get a more desired result!

Non-Parametric Segmentation

Instead of fitting an analytic function to the histogram of an RGB image, we can use the histogram itself to decide whether a pixel fits the ROI or not. We use histogram back-projection to essentially make a lookup table where a pixel gets a value according to its histogram value in NCC space.

We obtain a 2D histogram by binning r, g values converted to integers in to a matrix. This matrix is then concatenated to a look-up table using histogram backprojection, with indices being a function of the bin indices r, g .

Shown in Fig 5. is a result of segmenting Cuphead's shorts using different number of bins. We see that a larger amount of bins makes the lookup table more restrictive. Using 128 bins, even other red portions of in the image such as Mugman's tongue where not detected at all. On the other hand, 8 bins yielded artifacts around each character's outlines. The result does not depend on the PDF assumed like in the previous example, but it depends on the number of bins chosen by the user. In this example, the sweet spot is at about 32 bins. How did I get that number? Trial and error, all I wanted was Cuphead's shorts!

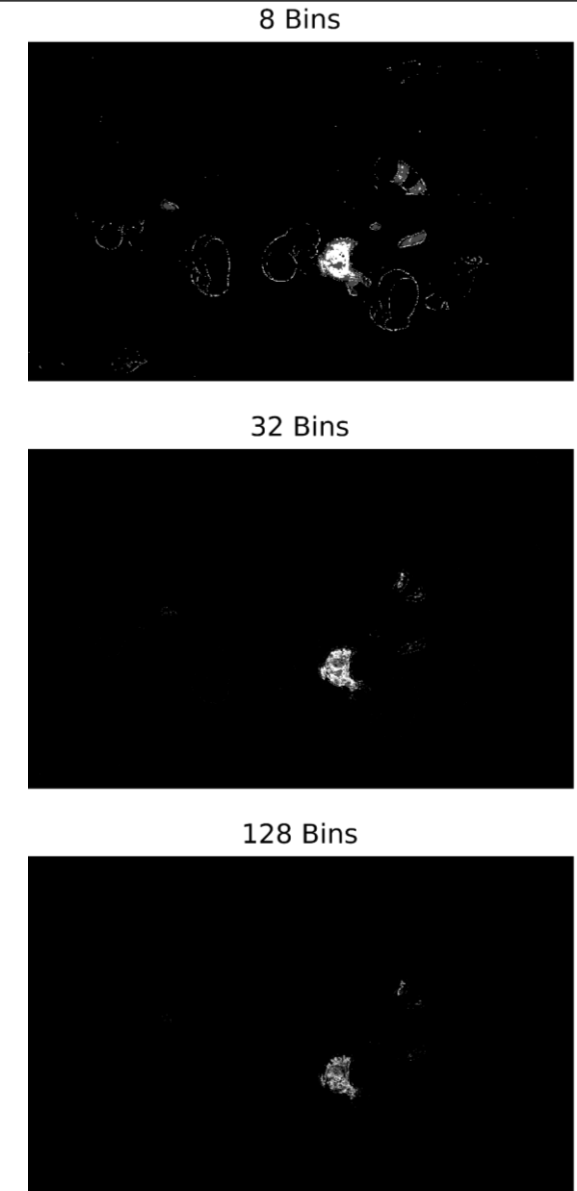


Fig 5. Non-parametric segmentation with varying bin quantities

Parametric vs. Non-parametric

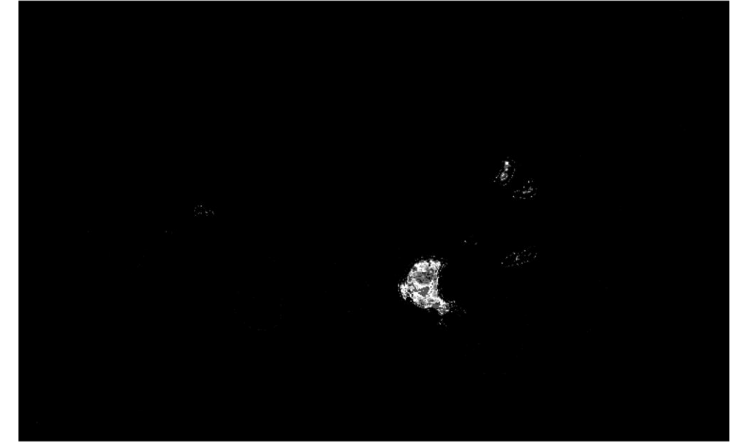
Original RGB Image



Parametric



Non-Parametric



Let's take a moment to compare these two methods of segmenting colored images. One involves solving an analytic probability function and one involves a simple look up table. Hence when it comes to computational efficiency, non-parametric segmentation uses less time and power. However computers these days can do parametric segmentation with ease so this is not a main concern.

Non-parametric segmentation is a more rigid algorithm that gives you what you want without complications. However, your ability to choose any reasonable analytic function for parametric segmentation gives it merit when it comes to versatility. In summary, both tools are useful and it just comes down to what you are going to use it for.

Key Takeaways

- If a Region of Interest (ROI) is distinct in the grayscale of an image, you can isolate it with segmentation by thresholding
- Otherwise you will need to do either a parametric or non-parametric probability distribution estimation. The former involves overlaying an analytic function to your image that will act as probabilities while the other involves histogram backprojection

Reflection

This course has become my favorite for this semester because even though it's challenging, I can make stuff with my own hands that isn't a problem set or some laboratory report scribbled on paper. Making my first project involving machine vision was fun and maybe I'll appreciate it more when time comes - but today I'm just burned out from Quantum 2 and Electromagnetism 2. Parametric segmentation came quite naturally to me and it was a fun Eureka moment when it clicked but Non-parametric segmentation was much harder for me. Luckily, a classmate [3] stepped in and taught me how I should code the algorithm. I helped him fix his code for canvas weave removal so it's nice to see collaboration between peers.

I did all required tasks, and created functions that can be applied regardless of the image and ROI fed to it so I think I deserve a 100/100. I'm really confident in my code for this activity! No bugs and it accepts all inputs with no errors!

References

- [1] Soriano, M. (2023). Feature Extraction Part 1 of 3: Image Segmentation. App Physics 157. National Institute of Physics, UP - Diliman.
- [2] NCC Space Image: https://en.wikipedia.org/wiki/Rg_chromaticity#/media/File:Rg_normalized_color_coordinates.png
- [3] Histogram Backprojection Code Outline: My Classmate, Nino Ramones